

# Why Sybil (sybild) CANNOT run as a remote Active Directory client

sybild impersonates by FORGING a TGT signed with the realm krbtgt key, which it reads from the KDC database via kadm5— AD exposes no such interface

## WINDOWS • Active Directory

### Active Directory Domain Controller — KDC

- issues tickets via the Kerberos protocol, but...
- exposes NO kadm5 / KDB key interface to clients
- the krbtgt key never leaves the domain controller
- pulling principal keys out of AD = DCSync
  - = full domain compromise (an attack, not an API)
- S4U is supported — but that is a CLIENT protocol, not the key-level DB access sybild requires

## LINUX • sybild (NVIDIA/sybil)

### sybild — impersonation mechanism

- ``sybil kinit user@REALM`` always FORGES a TGT
  - `krb::Credentials::forge(user, "krbtgt/REALM")` [lib.rs:266]
- signs it with the realm krbtgt key, which it reads from the KDC DB: `kadm5_get_principal_keys()` [krbutil.c:159] — daemon runs `kadm5_init()` [:42]
- no S4U-only mode — forging is unconditional
- ⇒ needs KDB access + co-location with an MIT / FreeIPA KDC

sybild asks for the  
krbtgt key from the KDC DB

AD will never export it  
→ sybild cannot operate against AD

## What actually works against AD → krb-credd

krb-credd is a Kerberos S4U \*client\* of AD: it holds a sybil-style SERVICE keytab and calls S4U2Self / S4U2Proxy over the wire — AD issues every ticket and authorizes delegation via msDS-AllowedToDelegateTo. No KDB access, no key reading, no co-location with the KDC. Same goal as Sybil (impersonate batch users), DIFFERENT and AD-supported mechanism (constrained delegation, not forging). See docs/krb-credd-architecture.